Chenxu Bai

Undergraduate | School of EECS | Peking University
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Education

Peking University, Robotics Engineering (transferred to CS; currently a minor) **2023.09 – 2025.09**
Peking University, Computer Science and Technology **2025.09 – Present**

- **GPA:** 3.*** / 4.0
- **Coursework:** Fundamentals of AI, Software Design, Continuous-time Diffusion Process in ML Theory, Information Theory, Computational Photography (Grade: **100**), Data Structures and Algorithms (Grade: **100**), Programming in AI (Grade: **100**)

Experiences

Xiaomi Mimo Team | Research Internship | **2026.06 – Present**

PKU YuanGroup | Research Internship | **2026.06 – Present**
Supervisor: Prof. Li Yuan | Beijing, China

Camera Intelligence Lab, IDM/NERCVT | Research Internship | **2024.11 – 2026.04**
Supervisor: Prof. Boxin Shi | Beijing, China

- My research focuses on the intersection of **computer vision** and **computational photography**, with a particular emphasis on neuromorphic camera imaging, including event-based video reconstruction and all-in-one image restoration.
- My research outcomes include:
  - **AE2VID**: **co-first-author** paper accepted to **CVPR 2026** on event-based video reconstruction.
  - **EvAIR**: **third-author** paper under review on event-based all-in-one image restoration.

Prof. Mohan Chen Research Group | Undergraduate Research Project | **2023.11 – 2024.11**
AISI (AI for Science Institute, Beijing) | Internship | **2024.08 – 2024.11**
Supervisor: Prof. Mohan Chen | Beijing, China

- My work focuses on **scientific computing**, mainly for **ABACUS** (Atomic-orbital Based Ab-initio Computation at USTC), an **open-source** density functional theory (DFT) package (Github Starred: 255) for large-scale electronic-structure simulations from first principles.
- My contributions include:
  - Binding selected modules of ABACUS to Python ("**Pythonization** of ABACUS").
  - **Maintaining** and **refactoring** parts of the ABACUS source code.

Publications

*Equal contribution †Corresponding author

- **[CVPR 2026] Chenxu Bai***, Boyu Li*, Peiqi Duan†, Xinyu Zhou, Hanyue Lou, and Boxin Shi†. AE2VID: Event-based Video Reconstruction via Aperture Modulation. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2026.
- **[Technical Report]** Zongjian Li, Zhiyuan Yan, **Chenxu Bai**, Chen Chen, Haoxiang Sun, Shaodong Wang, Feize Wu, Shenghai Yuan, Bin Lin, Zheyuan Liu, Yuwei Niu, Li Yuan†. UniWorld-Design: From Pixel Generation to Layer-Native Design. arXiv: 2608.03971, 2026.

Honors & Awards

- Merit Student, Peking University (2025)
- Luohua Scholarship, Knovva Academy (China) Inc. & Peking University (2025)

Skills

- **English:** CET-4, CET-6
- **Programming Language:** C/C++, Python, MATLAB, Rust